

Indrani Sen

(206)458-2884 | indrani.sen1@gmail.com | Austin, TX | [LinkedIn](#) | H4 EAD (No sponsorship needed)

Results-driven Business Intelligence Engineer with 6+ years of experience **and a recent MS in Computer Science (Georgia Tech, 2025)**. Expert in designing automated reporting solutions, building ETL/ELT pipelines, and translating complex datasets into actionable business insights. Proven track record of delivering measurable impact through data-driven decision-making, with deep expertise in AWS cloud architecture, advanced SQL, and data visualization tools including QuickSight and Tableau

EDUCATION & CERTIFICATION

Georgia Institute of Technology | MS, Computer Science

Dec 2025

Kalinga Institute of Industrial Technology | MS, Computer Application

May 2014

AWS Certified Solutions Architect - Associate

Pursuing

PROFESSIONAL EXPERIENCE

Business Intelligence Engineer, Tata Consultancy Services | Dec 2014 – Jun 2021

- **Designed and delivered** automated reporting dashboards processing **10M+** daily records, enabling executive decision-making for utility operations serving **8+** million customers across multiple state organizations
- **Engineered high-performance ETL pipelines** processing millions of records daily, migrating from legacy line-by-line inserts to optimized bulk loading operations, reducing database write latency by **40%** and improving data freshness for business reporting
- **Developed comprehensive analytics solutions** using advanced SQL (Db2, MySQL, PostgreSQL, Oracle) achieving **40%** query performance improvement through indexing strategies and query optimization, enabling real-time insights from billion-row datasets
- **Built multi-threaded data processing frameworks** with data quality validation and automated alerting, achieving **25+** records/second throughput with **99.9%** accuracy and **100%** audit trail compliance for distributed infrastructure systems
- **Led requirements gathering sessions** with business stakeholders, translating complex business needs into technical analytics solutions and data models; presented data-driven insights to senior leadership through compelling visualizations
- **Established KPI tracking frameworks** and performance metrics for distributed systems, monitoring operational efficiency, system uptime (**99%**), and resource utilization; delivered analytics automation eliminating **300** manual hours weekly
- **Managed cross-functional collaboration** with engineering and business teams (**15+ members**) across **\$2M+** project budgets, using Agile methodologies (JIRA) to deliver analytics initiatives achieving **\$200K** annual cost savings and **12%** revenue growth

TECHNICAL SKILLS

- **Data Analytics & SQL:** Advanced SQL (Redshift, PostgreSQL, MySQL, Oracle, Db2), Query Optimization, Complex Joins, Window Functions, CTEs, Performance Tuning, Database Design
- **ETL/ELT & Data Pipelines:** Data Pipeline Architecture, Batch and Real-Time Processing, Data Quality Validation, Automated Workflows, Data Lineage, Multi-Threaded Processing

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- **Data Engineering & Analytics:** Distributed Systems, Data Warehousing, Dimensional Modeling, GenAI/LLM Integration, Statistical Analysis, A/B Testing
- **Programming & Development:** Python (Pandas, NumPy), Java, Spring Boot, RESTful APIs, FastAPI, Angular, React
- **Data Visualization & BI Tools:** Amazon QuickSight, Tableau, Dashboard Design, KPI Tracking, Self-Service Analytics, Data Modeling
- **AWS & Cloud Technologies:** AWS Solutions Architecture, S3, Redshift, Lambda, API Gateway, CloudWatch, EC2, DynamoDB, Bedrock, Cloud Migration

KEY TECHNICAL PROJECTS

GenAI Fairness Agent & RAG System

- **Developed analytics system** using **RAG architecture** and NLP pipelines to audit employee performance reviews against **EEOC legal guidelines** and mitigate litigation risks
- **Built NLP pipelines** using **Sentence Transformers** and **FAISS** to detect gender-coded language patterns (agentic vs. communal) and calculate Disparate Impact metrics in real-time
- **Designed an automated remediation workflow** that flags biased feedback and generates neutralized rewrites, preserving performance data while ensuring regulatory compliance

AWS Content Aggregation Platform

- **Developed enterprise-grade** serverless content processing system using Lambda, S3, Bedrock, and API Gateway with AI-powered categorization achieving 90%+ accuracy
- **Built scalable data warehouse** supporting complex analytical queries with 3-second response times for 1000+ concurrent users
- **Implemented real-time monitoring** through CloudWatch dashboards with automated quality scoring and data validation workflows.

GenAI-Powered Healthcare Risk Platform

- **Architected a provider-facing dashboard** using **FastAPI** (backend) and **React** (frontend), processing **FHIR-compliant** JSON patient data to calculate clinical risk scores
- **Integrated Google Gemini LLM** to transform Framingham risk assessments into personalized, evidence-based patient education and lifestyle recommendations
- **Managed end-to-end delivery** including **Docker** containerization for consistent deployment and the generation of synthetic datasets to test high-risk patient scenarios

PUBLICATIONS

"Path to Personalization: A Systematic Review of GenAI in Engineering Education" (KDD 2024)
Co-authored systematic review of 162 GenAI studies (KDD 2024); **led global team of 6 researchers** to define taxonomy and crowdsource content analysis.